

Dr. Tarun Teja Kondraju

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Profile

Ph.D. in Spatial Informatics with interdisciplinary research experience in remote sensing applications for water, agriculture, and environmental systems. My expertise lies in integrating satellite and UAV observations with machine learning and cloud computing to model water quality, ecosystem health, and crop dynamics across diverse climatic and geographic regions. I maintain a strong record of peer-reviewed publications and am motivated to advance transferable, reproducible remote sensing methodologies that support sustainable resource management and climate resilience.

Academics

Ph.D. in Spatial Informatics

International Institute of Information Technology, Hyderabad | 2013–2021 | GPA: 7.67

Thesis: Water Quality in Inland Water Bodies and Its Interaction with Land Use

M.Tech. in Energy and Environmental Engineering

Vellore Institute of Technology | 2010 – 2012 | GPA: 7.68

B.Tech. in Biotechnology

St. Martins Engineering College, Hyderabad | 2006 – 2010 | 62.85 %

Work Experience

Indian Agricultural Research Institute – ICAR (Feb 2022 – Present)

1. Working on Google Earth Engine-based real-time monitoring application for monitoring crop health.

AvakAza Geoscience and Research Technologies (Oct-2020 to Oct- 2021-PT)

2. Worked as a GIS research consultant and trainer
 - Took online training classes on the application of ArcGIS and QGIS for various applications.
 - Trained professionals and students from various backgrounds.

IIT-Hyderabad (Jan-2013 to Nov-2021)

1. Senior Research Fellow (Nov-2021 to Feb-2022)

- DST project for satellite-based water quality monitoring in Bhadra and Tungabhadra Reservoir

2. Research Experience (Ph.D.) (Jan-2013 to Nov- 2021)

3. Research Assistant

- Western Ghats-APN Project under OSGEO organization

- Worked towards the Capacity Building of faculty members from various institutions across India.

4. Teaching Assistant

- Took several classes for undergraduate, graduate, and research students
- Conducted tutorials on
- Applying Remote Sensing data for various studies.
- Vector and Raster data processing
- Application of GIS, Geo-Servers, and Web-GIS.
- GIS software for performing various spatial analysis

Research Interest

Use remote sensing data to

2. Enhance agricultural productivity in different climates and regions.
3. Improve early detection of algal blooms in freshwater lakes to improve fishing and water quality.
4. Address various issues for the improvement of the healthy climate and environment.
5. Develop various tools that would allow users to make quick and practical decisions.

Acquired Skills

Technical

1. Processing Various optical satellite datasets
2. Extracting water quality and crop health information using remote sensing.
3. Generating deliverables from the remote sensing data using spatial analysis techniques.

GIS Software

1. Online Data Processing, visualization, and app building using the Google Earth Engine,
2. DroneDataProcessing: Pix4D, and WebODM
3. Lidar Data Processing: Micro-Station
4. GIS functions: ArcGIS, and QGIS
5. Image processing software: ENVI

Research Projects

Post-Doctoral Research: Crop health assessment using hyperspectral, multispectral, RGB and thermal sensors data collected from UAVs and Satellite platforms by applying machine learning, physical modelling and also cloud computing.

Ph.D. Research: Water quality modeling in Nagarjuna Sagar Reservoir, using Sentinel 2A/B satellite data to track nutrient contamination.

Coral Reef Study: Post-tsunami damage assessment of coral reefs near Andaman Islands using Landsat data.

Urban Waste-to-Energy Project: Estimation of electricity generation potential from organic waste in Hyderabad.

Products Developed

PUSAeCMS : A Google Earth Engine-based application for efficient big data processing for proving near real-time crop health management using vegetation indices, water indices, engineered composite indices, physical process – based biophysical variables estimation. The application allows the user to perform quantitative analysis and visualize temporal changes occurring in crop health at any given location.

VASUDHA: An application to monitor soil health using by N, P, K, OC, EC, and pH estimated from VNIR/SWIR and MIR spectra generated from spectroradiometer and FTIR respectively in real-time. The application can also be used in offline mode to assess soil quality from large quantity of spectral data in an instance.

PUSAMIA: An application to co-register, layer stack and assess proximal multispectral images of crops.

PAUDHA: An application to assess crop health from large amounts of data provided by spectroradiometer in real-time or offline mode.

Publications

2025

Kondraju, T. T., Sahoo, R. N., Rejith, R. G., Bhandari, A., Ranjan, R., Reddy, D. V. S. C., & Ramalingam, S. (2025). Retrieving canopy chlorophyll content from Sentinel-2 imagery using Google Earth Engine. *Remote Sensing*.

Rejith, R. G., Sahoo, R. N., Kondraju, T. T., Bhandari, A., Ranjan, R., & Moursy, A. (2025). Generation of synthetic hyperspectral image cube for mapping soil organic carbon using proximal remote sensing. *Remote Sensing*.

Rejith, R. G., Sahoo, R. N., Kondraju, T. T., Bhandari, A., & Ranjan, R. (2025). A hybrid machine learning approach for monitoring wheat crop traits using proximal hyperspectral remote sensing. *Remote Sensing*.

Sahoo, R. N., Rejith, R. G., Kondraju, T. T., Ranjan, R., Bhandari, A., Gakhar, S., Asim, M., Verrelst, J., Kaur, R., & Singh, T. (2025). Scaling-up plant chlorophyll retrieval from proximal to UAV-borne hyperspectral data using a Gaussian process hybrid model. *International Journal of Remote Sensing*, 1–25.

Reddy, D. V. S. C., Sahoo, R. N., Kondraju, T. T., Rejith, R. G., Ranjan, R., Bhandari, A., Moursy, A., Tripathi, S. C., & Kumar, N. (2025). Drone-based multispectral imaging for precision monitoring of crop growth variables. *Biology and Life Sciences Forum*, 41(1), 10.

Rejith, R. G., Sahoo, R. N., Ranjan, R., Kondraju, T. T., Bhandari, A., & Gakhar, S. (2025). Estimating leaf area index of wheat using UAV-hyperspectral remote sensing and machine learning. *Biology and Life Sciences Forum*, 41(1), 11.

2024

Latwal, A., Kondraju, T. T., Rehana, S., & Rajan, K. S. (2024). Examining chlorophyll-a concentrations in tropical reservoirs under various land-use changes using Sentinel-2 and Google Earth Engine: Bhadra and Tunga-Bhadra, India. *Journal of Contaminant Hydrology*, 265, 104388.

Sahoo, R. N., Kondraju, T. T., Rejith, R. G., Verrelst, J., Ranjan, R., Gakhar, S., & Chinnusamy, V. (2024). Monitoring cropland leaf area index using Gaussian process regression and Sentinel-2 surface reflectance data in Google Earth Engine. *International Journal of Remote Sensing*, 45(15), 5008–5027.

Sahoo, R. N., Rejith, R. G., Gakhar, S., Verrelst, J., Ranjan, R., Kondraju, T. T., & Chinnusamy, V. (2024). Estimation of wheat biophysical variables through UAV hyperspectral remote sensing using machine learning and radiative transfer models. *Computers and Electronics in Agriculture*, 221, 108942.

Reddy, D. V. S. C., Sahoo, R. N., Kondraju, T. T., Rejith, R. G., Ranjan, R., Bhandari, A., Moursy, A., Tripathi, S. C., & Kumar, N. (2024). Drone-based multispectral imaging for precision monitoring of crop growth variables. *Drones*.

Kondraju, T. T., & Sundara, R. K. (2024). Studying the impact of precipitation on nutrient loading into the Nagarjuna Sagar Reservoir from its contributing watershed using the SWAT model.

2023

Sahoo, R. N., Gakhar, S., Rejith, R. G., Verrelst, J., Ranjan, R., Kondraju, T. T., Meena, M. C., Mukherjee, J., Daas, A., & Kumar, S. (2023). Optimizing the retrieval of wheat crop traits from UAV-borne hyperspectral imaging with radiative transfer modelling using Gaussian process regression. *Remote Sensing*, 15(23), 5496.

Gakhar, S., Sahoo, R. N., Rejith, R. G., Kondraju, T. T., & Ranjan, R. (2023). UAV sensing of rice for precision nitrogen monitoring. *International Rice Research Institute Conference Proceedings*.

Rejith, R. G., Sahoo, R. N., Verrelst, J., Ranjan, R., Gakhar, S., Anand, A., & Kondraju, T. T. (2023). UAV-based retrieval of wheat canopy chlorophyll content using a hybrid machine learning approach. In *Proceedings of the IEEE India Geoscience and Remote Sensing Symposium (InGARSS)* (pp. 1–4). IEEE.

2022

Kondraju, T. T., Mandla, V. R., Chokkavarapu, N., & Peddinti, V. S. S. (2022). A comparative study of atmospheric and water column correction using various algorithms on Landsat imagery to identify coral reefs. *Regional Studies in Marine Science*, 49, 102082.

Manoharan, S., & Kondraju, T. T. (2022). Impact of Bengaluru urban growth on its lakes. In *Proceedings of the IEEE Mediterranean and Middle-East Geoscience and Remote Sensing Symposium (M2GARSS)* (pp. 138–141). IEEE.

2021

Kondraju, T. T., & Rajan, K. S. (2021). Understanding the impact of fertilizer application on Nagarjuna Sagar Reservoir. In *Proceedings of the HYDRO 2021 International Conference*.

Kondraju, T. T., Krishna, A. V., & Mandla, V. R. (2021). Evaluation and comparison of evapotranspiration losses using various algorithms. In *Proceedings of the 2nd International Conference on Sciences, Engineering and Technology (SET)*.

2019

Kondraju, T. T., & Rajan, K. S. (2019). Water quality in inland water bodies: Hostage to the intensification of anthropogenic land uses. *Journal of the Indian Society of Remote Sensing*, 47(11), 1865–1874.

Kondraju, T. T., & Rajan, K. S. (2019). Excessive fertilizer usage drives agriculture growth but depletes water quality. *ISPRS Annals of Photogrammetry, Remote Sensing and Spatial Information Sciences*, IV-3/W1, 17–23.

2016

Kondraju, T. T., & Rajan, K. S. (2016). Understanding the behavior of contamination spread in Nagarjuna Sagar Reservoir using temporal Landsat data. *International Archives of Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLI-B8, 343–348.

2014

Kondraju, T., Mandla, V. R. B., Mahendra, R. S., & Kumar, T. S. (2014). Evaluation of various image classification techniques on Landsat to identify coral reefs. *Geomatics, Natural Hazards and Risk*, 5(2), 173–184.

2012

Teja, T., Mandla, V. R., & Krishna, A. V. (2012). Comparative study of evapotranspiration losses using various algorithms. *International Journal of Earth Sciences and Engineering*, 5(6), 1683–1691.

Tarun Teja, K., Srinivasa Kumar, T., Mandla, V. R., & Mahendra, R. S. (2012). A comparative study of atmospheric correction algorithms on Landsat imageries for identifying coral reefs. In *Proceedings of the National Seminar on Coastal and Island Ecosystems*.

2011

Samui, P., Mandla, V. R., Krishna, A. V., & Teja, T. (2011). Prediction of rainfall using support vector machine and relevance vector machine. *Journal of Earth Sciences India*, 4(4), 188–200.

Workshops / Trainings Attended

International Workshops & Training Programs

1. **International Training Workshop on “Principles of Ocean Remote Sensing and Its Applications”**
International Training Center for Operational Oceanography (ITCOcean) & Indian National Centre for Ocean Information Services (INCOIS)
26–30 July 2021, Hyderabad, India.
2. **International Training on “Training, Validation, and Application of Remote Sensing Products in Ecology and Environment for BRICS Countries”**
Organized by leading Remote Sensing Institutions of BRICS Nations
24–26 May 2022.

Invited Resource Person / Expert Lectures (Workshops Attended as Speaker)

3. **Training Programme on “Drone Technology & Its Applications in Agriculture”**
Organized for Scientists, Academicians, Research Scholars, and PG/PhD Students
2–3 December 2024
Role: *Invited Resource Person*
Contribution: Delivered a lecture and submitted an 8-page training manual write-up.
4. **Training-cum-Workshop on “Drone Technology and Its Application”**
Jointly organized by ICAR-CIFRI, Barrackpore, Kolkata & ICAR-IARI, New Delhi
11 December 2024
Role: *Invited Speaker*
Contribution: Delivered a lecture on drone-based imaging spectrometry and precision agriculture applications.
5. **Guest Lecture on “UAV Mission Planning, Data Acquisition (RGB, multispectral, hyperspectral and thermal), and Analytics for Precision Agriculture”**
18th December, 2025.
Role: *Invited Speaker.*
Contribution: Delivered a lecture on drone-based imaging spectrometry and precision agriculture applications.

References

Dr. Rabi N Sahoo

Principle Scientist

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